



Classifying Routing Algorithms upon Clustered Based Wireless Sensor Networks: A Survey

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ABSTRACT

The previous couple of years have encouraged enthusiasm among interested researchers in the field of clustered based algorithms upon Wireless Sensor Networks (WSNs). This is for many reasons; such as more scalability and efficient in energy that outperform other existing algorithms in literature. The clustered based protocols must meet some inherited requirements in sensor devices, like limited energy resources, restricted processor capability and limited coverage area in both states of underlying algorithm round of work. In this context, the current survey classified the existing algorithms in literature based on objectives and functions to address the weaknesses in clustered based algorithms work including stage of election of cluster heads and super cluster heads, formation stage of clusters, aggregation stage of data and communication stage of data. Also this survey included all implemented techniques and approaches carried out in literature in this classification. Furthermore, all mentioned advantages and disadvantages within any approach have been introduced in more details, while classifying algorithms work for each stage has given to offer more profound comprehension of clustered based algorithms in literature. At last and in light of this survey, all issues and proposed taxonomy based on used methods with brief comparison for popular clustered based algorithms have been summarized. Finally, proposed future works in underlying field of study that can be further debated have been introduced as well.

1. INTRODUCTION

Recently, there have been many studies introducing the capabilities of WSNs in many scenarios, like home and environmental monitoring, battlefield monitoring and habitat surveillance. Generally, each sensor device is supplied by a limited battery as a power source, and is considered as unattended device after

deployment. Based on these inherited limitations in sensor node, majority of existing researches in the field of WSNs study the challenges relating to Quality of Service (QoS) of underlying scenarios like sensor network functionality as real time, network scalability and secured data transmission. In this context, the inherited limited capabilities within WSNs,

such as battery storage, low processing, small memory size, and communication range challenges should be investigated well. Besides, the dynamic topology changes and sensor nodes demises need insight and more investigation too.

The cluster based protocols are considered as the most compatible protocols with WSNs field for many advantages, such as distributing the energy load among nodes, reducing transmission power and bandwidth demands, avoiding the redundant data in aggregation process stage, avoiding data loss within steady data transmission phase, and providing more scalable WSNs network.

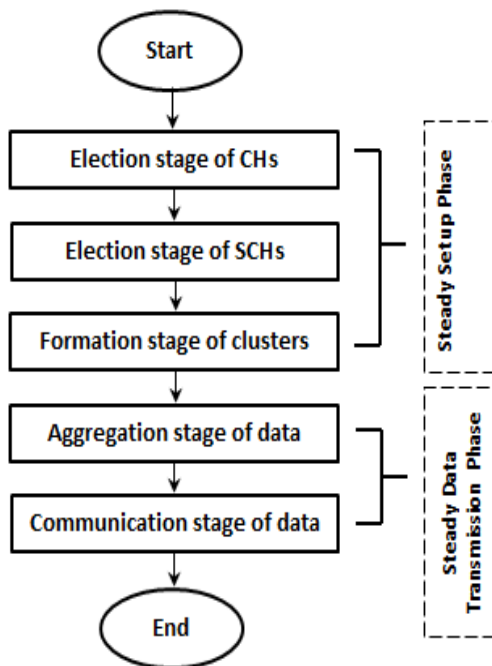


Figure (1): The flow chart of one complete cycle for cluster based protocols

To the best of our knowledge, wireless sensor devices in cluster based topology can be divided into five groups and the brief description of these groups is shown in Table 1. However, Fig. 1 shows the flow chart of one complete cycle for cluster based routing protocols process over WSNs.

The recent approach in (Singh and Sharma 2015) introduced a survey on cluster based

WSNs. They proposed a taxonomy of cluster based routing protocols. This survey is not comprehensive due to the lack of knowledge about recent network topology approach used in (Gawdan 2014; Gawdan and Sarhan 2016) like the election of Super Cluster Head (SCHs) nodes which are responsible to relay the collected data from CH nodes towards the destination node which is here Base Station (BS).

The current work focuses on group of challenging factors associated to designing routing algorithms as shown in Table 2. Also this work focuses on classifying cluster based algorithms according to the implemented methods in each phase of routing works. Furthermore, this work will be helpful for researchers those are interested in modification, development and optimization of routing protocols for cluster based WSNs.

The organization of the current survey is as follows: The literature review is presented in Section 2. The election of super cluster head and cluster head nodes is introduced in Section 3 and Section 4, respectively. In Section 5, the formation stage of clusters scenarios is presented. Section 6 surveys aggregation stage of sensed data in insight. In Section 7, various scenarios in the communication stage of data are surveyed. Section 8 proposes a taxonomy for popular cluster based algorithms and summarizes a brief comparison between popular existing algorithms on the basis of well-known factors in the field of WSNs. Finally, in Section 9, we outline the issues and solutions for cluster based approaches, and then some future research areas for the design are proposed as well.

2. LITERATURE REVIEW

Clustered based algorithms enhanced the network performance of WSNs in terms of latency, throughput and energy consumption.

Algorithms including rotating task of SCHs and CHs sensor nodes are essential and noticeable in term of balanced energy distribution. Most recent existing cluster based researches in literature follow the existing common setup phase for WSNs functionality. This common setup phase is composed of election of CHs and formation stage of clusters.

Table 1. Groups of wireless sensor devices within cluster based topology

No.	Node Group Name	Description
1	Super Cluster Head (SCH)	Elected and placed within the boundary of entire network close to Base Station (BS).Collecting the merged data coming from Cluster Head (CH) nodes and sending or relaying these data towards sink node BS node.
2	Cluster Head (CH)	Selected as head node within specific cluster boundaryand then collecting the detected data from normal nodes within cluster.Sending the merged data to next hop node (may be SCH or BS which node is the nearest).
3	Base Station (BS)	This node is considered as powerful node with given more capable processing. This is considered as sink node within entire network where all the sent data from network are processed based on underlying application scenario.
4	Relay Node (RN)	This node is used in multi hop scenario. Its duty for relaying sent data from other sensor nodes to the BS.
5	Normal Node (NN)	The normal nodes are considered as the majority of nodes within entire network. They provide the sensed data towards their SCH or CHs based on their locations and the underlying application.

The approach in (Gawdan 2014; Gawdan and Sarhan 2016) adds new stage named election stage of SCH nodes to the existing setup phase in literature. The SCH node is elected among elected CHs nodes based on their distances from BS and residual energy. Besides, the new optimal number for CH nodes has been recorded in this work which is 7% of the active nodes. Furthermore, by choosing this optimal number by this work carried out saving of 2% of total energy expenditure within network. To this end, the entire network lifetime is thus elongated to some extent.

The approach in (Singh and Sharma 2015) introduced a survey on cluster based WSNs. They proposed a taxonomy of cluster based routing protocols. They also surveyed the status of research and classified the different clustering method. To the best of our knowledge, this work is not comprehensive, as they do not update their work to the recent approach used as in (Gawdan 2014; Gawdan and Sarhan 2016). Many recent works have been conducted to overcome the drawbacks of cluster based routing algorithms but there are still many issues such as, scalability, fault tolerance, redundancy and others need to be discussed toward improving the functionality of methods used in this field of study (Sikanderet *al.*, 2013; Kumar 2014).

The past four years in literature have been witnessed few existing surveys relating the field of clustered based algorithms over WSNs as in (Singh *et al.*, 2010; Li *et al.*, 2011). References (Singh *et al.*, 2010; Boyinbodeet *al.*, 2011) has been provided survey on the durability and shortcomings of small numbers of existing clustered based routing algorithms in a separate manner, no longer mentioning any classification upon them. Reference (Li *et al.*, 2011) presents a deep understanding into clustered based algorithms implemented to execute functionality upon large size networks.

Based on energy efficiency consideration as hot issue in large size WSNs, the current survey groups the existing clustering algorithms into overheads reduction, consumed energy mitigation and balanced energy distribution.

Based on our knowledge, all above mentioned surveys in literature outline a comparative study for their attributes and leaving the significant limitations that exist in each phase of cluster based protocols for WSNs. So far, this current survey is the most recent and comprehensive survey that provides deep

understanding and investigates most of recent existing clustering protocols for WSNs. This is through providing classification toward their contributions and dealing their drawbacks for individual phases of clustered based algorithms. Furthermore, the survey in this study provides guideline as essential point to guide researchers who are needed to follow up research in this field.

3. ELECTION STAGE OF SUPER CLUSTER HEAD NODES

Due to the important of the election stage of SCH nodes and keeping the common setup phase followed in most recent existing researches in literature, we present this separate section for surveying the election stage of SCH nodes. While the existing common setup phase will be surveyed throughout the next sections.

The approach in (Gawdan 2014; Gawdan and Sarhan 2016) provides a novel approach for election of SCH nodes. After the election of CH nodes, they broadcast their locations, Residual Energy (RE) and Identifiers (ID) to the BS. Then, the base station computed average energy (E_A) and distances D_{CH-BS} to the mentioned CH nodes. Consequently, the CH nodes are elected to become SCH nodes when the conditions given as in (1) and (2) are valid.

$$E_{SCH} \geq E_A \quad (1)$$

$$D_{SCH-BS} \leq D_{CH-BS} \quad (2)$$

Here E_{SCH} stands for candidate SCH nodes energy, the D_{SCH-BS} stand for the distances of the candidate SCH node to BS while D_{CH-BS} stand for the distances of CH nodes from the BS. By using this method, the power will be balanced within the entire network. To this end, the task of BS is to send "SCHs" packet including identifiers ID_{SCH} of SCH nodes. After matching the ID_{SCH} then the sensor node will become a SCH node else, it remains CH node.

It is obvious that this scheme provides fewer packet overheads, latency interval, and consumed energy, and it out-performs the existing approaches in literature like (Sharma and Ghose 2011; Karlofet *al.*, 2004; Luket *al.*, 2007). Besides, it adds very small increase over TinyOS stack of (Karlofet *al.*, 2004) not more than 5.6 %.

4. ELECTION STAGE OF CLUSTER HEAD NODES

4.1 Self Arranged Based Approaches

All existing self-arranged based approaches in literature have been witnessed that the election of CH nodes decision is distributed in entire WSNs. The approach in (Gawdan 2014; Gawdan and Sarhan 2016) introduced the election of Cluster Head (CH) nodes. The election of CHs is started by implementing of Trust Cluster Head Mechanism (TCHM) technique to provide Trust Level (TL) for every node within its cluster. Consequently, the eligible CH node broadcast "Adv" message containing CH's Identifier (ID), the Normal Node (NN) can join to the selected CH which requires minimum transmission energy by sending "Join-Req" message to the required CH. Then the elected CHs broadcast Time Division Multiplexer (TDM) schedule "Schedule" message containing the slot time for every node within particular cluster. In each round we choose different number of CHs in the network on the basis of number of alive nodes. The optimal number of CHs is 5% of the total alive nodes in the network for every new round as done in LEACH (Heinzelman *al.*, 2000; Heinzelman *al.*, 2002). Each CH sends its ID, residual energy and location info to the BS for furthermore processing.

Reference (Heinzelman *al.*, 2000) was the first self-arranging clustered based algorithm addressing the issues of WSNs. The algorithm adopted the approach of randomization to achieve evenly balanced energy distribution

among sensor devices in WSNs. Rule here is that, each sensor node establishes arbitrary number between (0, 1) and comparing it with a calculated $T(n)$ value as in (3). To this end, the node assures its situation to become CH sensor node in underlying round or not. To this context, the sensor node becomes a CH node when the established chosen number is less than calculated value $T(n)$, and then selected CH advertises the “CHs” message to all member in WSNs. All members nodes must connect to the cluster within received signal strength. Here P_{CH} represents desired percentage of CHs and r_C represents underlying round. While n represents the number of sensor nodes which did not become CHs nodes within previous $1/P_{CH}$ rounds (Heinzelman *et al.*, 2000).

$$T(n) = \frac{P_{CH}}{1 - P_{CH} * (r_C \bmod 1/P_{CH})} \quad (3)$$

The scheme in (Peng and Zhang 2010) presents an updated calculated threshold to enhance the election of CHs by computing the rate of current to the initial energy E_C/E_i relating to sensor nodes. Based on (4), the algorithm provides a solution by mitigating the issue of probability chance reduction of individuals which lead to exhausting their energy power:

$$T'(n) = \frac{P_{CH}}{1 - P_{CH} * (r_C \bmod 1/P_{CH})} * \sqrt{\frac{E_C}{E_i}} \quad (4)$$

Reference (Zhang *et al.*, 2010) introduced a new approach considered a weighted residual energy, node distribution density and distance among sensor devices and the BS. In this algorithm, the modified threshold value enhances the method used for CH election by LEACH (Heinzelman *et al.*, 2000) to multilevel decision making scenarios. The selection of suitable ratio weights is significant to generate efficient threshold that does not reduce the sensor network performance.

Reference (Penget *et al.*, 2010) presents balanced energy distribution algorithm, which handles asymmetric data throughput in coal mine. This algorithm assumed the CHs nodes which are nearer to the sink BS exhaust their energies very soon. To handle this issue, the algorithm introduces a CH node election approach where the closet areas to the BS elect more CHs nodes than the further areas. Based on this context, the sensor nodes possessing the higher residual energy and closest to destination BS gain more guarantee to become CH nodes. To this end, the CHs nodes density follows the effective deployment density for backbone of network.

To the best of our knowledge, in distributed scenarios, discovering the neighboring sensor nodes and collecting resource information slows the setup phase work and adds additional overhead by equipping the sensor nodes with Global Positioning Systems (GPS) technique. Besides, the decision making built on GPS does not offer the optimized elected CH node from a perspective of network scale.

4.2 Aided Based Approaches

To overcome the shortcomings imposed by previous distributed approaches about the energy expenditure and the fair displacement of CH nodes within entire WSNs, the aided based approaches such as BS and CH nodes are proposed to meet the restricted resources of WSNs.

The Base Station Aided Based Approaches:

The LEACH-Centralized(LEACH-C) (Heinzelman *et al.*, 2002) provides fair solutions for issues imposed by LEACH (Heinzelman *et al.*, 2000). In (Heinzelman *et al.*, 2002), all sensor nodes broadcast their locations and residual energy levels to the sink node which is in our case is BS, and then the BS computes average energy level of the entire WSNs. To this end, sensor nodes possessing energy above the computed value have more chances to be CHs for the current round.

Reference (Hosseinalipouret *al.*, 2010) proposes a new algorithm. The election stage of CHs is based on the calculated distances of sensor devices with others. In this respect, the entire WSN is divided into several equal areas according to the distances and positions of devices. The nodes probability for election stage of CHs based on rate of residual energy and the distance is determined by BS. This algorithm utilizes two predefined positions of sensor nodes with the help of sink BS to find the positions of sensor nodes for entire WSN. The BS generates distances table which can be updated with respect to determined data provided by the two preselected nodes. Reference (Wang *et al.*, 2007) mentions variation of optimum numbers of CH nodes per each round within clustered based algorithms processes according to the topology changes. In this algorithm, the CHs nodes transmit their combined messages including the size of their clusters to the BS. This leads to compute the optimum percentage number of CHs nodes for the next round based on alive sensor nodes within entire network. Factor of optimum percentage number of CH sensor nodes must be topology aware. This helps to assure network efficiency, network robustness and scalability of the algorithm to the topology changes.

The Cluster Head Nodes Aided Based Approaches: Within re-clustering process, extra overheads are added to the WSNs. These overheads are imposed through extra broadcastings of control messages within entire network. Therefore, some approaches are introduced as in (Nazir and Hasbullah 2010; Zhang *et al.*, 2011). These solutions help to mitigate the drawbacks of re-clustering process and exploiting the advantages of round-based clustering algorithms.

To enhance WSN performance and to reduce extra energy overhead during re clustering processes, the solutions in (Haoet *al.*, 2010; Zhang *et al.*, 2011) implement redundant and backup CHs nodes. In this respect, electing two nodes possessing highest residual energy as potential CH nodes in first round of setup phase. Also one of them is

assumed as premier CH sensor node while the second as the redundant CH sensor node are conducted in (Haoet *al.*, 2010). Based on this algorithm, sensors devices are ranked according to amount of residual power and hops number from others. The first rank assigned sensor node selected as main CH while assigned next position sensor considered as redundant CH node. CH sensor device which is redundant dynamically updates its self with the cluster information from main CH node. This helps to alternate the main CH node when any failure might happen in case of comprising or energy exhaustion.

4.3 Cross-Parameter Evaluation Based Approaches

The election stage of CH nodes is considered as main stage in clustering algorithms. This is true, because of its effects on other coming stages within protocol process. To ensure the correct election stage of CH node, the election must be based on cross-parameter decision issues among entire WSNs. One of the most popular existing solutions for election of CH nodes using cross-parameter evaluation based approaches is Fuzzy Logic Controller (FLC). Fuzzy logic controller is defined as a problem solving control system which offers the easiest way to access the definite debate depended on illustrative language for handling entered data as a human operator.

Reference (Torghabehet *al.*, 2010) proposes two-level fuzzy decision making that offers the regional and global levels. In this respect, the paper concentrated on some physical aspects related to sensor devices like battery power and degree of sensor node in the local view. For global view, factors of network scale are involved for conducting optimized energy overhead. The sensor nodes number qualified in local level via utilizing residual energy and existing number of neighboring sensor devices which are considered inputs to the regional fuzzy scheme. The elected nominees are computed again in global level according to

factors, such as centrality, nearness from sink node which is BS and distance among CHs nodes. Based on simulation results, this approach outperforms LEACH in lifetime, remaining power and energy expenditure deference per cycle. Based on our knowledge, the complication and added overheads of two-level FL are no longer studied well by this approach.

Table 2. Group of challenging factors associated to designing routing algorithms

Challenging Factors	Brief description
Energy consumption	Based on limited resources of WSNs, the routing protocols must be capable to accommodate the trade-off between energy consumption and accuracy.
Node deployment	This factor has more effects on the performance of underlying routing protocol. When the underlying application is regarded to event detection, it is recommended to have a random node deployment to achieve interested goals rather than manual one.
Nature of node	The most existing works in literature assumes that the sensor devices are stationary; however, there is a need to mobility of both normal node and base station in recent applications.
Scalability	In massive amount of sensor nodes within underlying network networks, it is very hard for each node to maintain global information upon underlying network topology.
Quality of service (QoS):	Some QoS parameters are bandwidth, latency and throughput. Some applications like target detection and tracking applications demand low latency for time sensitive data, while multimedia application needs high throughput.
Application	In general, sensor nodes use various methods such as, time driven, query driven and event driven methods for collecting interested data from environment. The underlying application determines the required method within the entire network to achieve the task.
Coverage	This factor is an essential design issues. A given node's view of the environment is limited in accuracy and in range.

5. FORMATION STAGE OF CLUSTERS BASED APPROACHES

Formation stage of clusters is started after the election stages of CH / SCH nodes are over.

The elected CHs broadcast their “CHs” messages to announce their election to neighboring nodes, and then each sensor node sends back a “join-message” to elected CH sensor node within coverage range. This section introduces optimal clustering and fault management based approaches:

5.1 Optimal Clustering Based Approaches

Many CH elected based approaches in literature agree the formation stage of clusters algorithm conducted by LEACH protocol and they provide some solutions related to the clustering issues in election of CH phase. While others such as in (Peng and Zhang 2010; Gou and Yoo 2010; Jiaet al., 2010) present resource-aware approaches for formation stage of clusters in a separate way.

Reference (Peng and Zhang 2010) introduces the relationship between size of cluster and the distance to the sink node which is in our case BS. The size of cluster is addressed to provide balanced energy consumption within entire WSNs based on number of hops to BS. The clusters size is ranged according to linear relation with distances of sensor nodes to the sink node.

Reference (Gou and Yoo 2010) presents a centralized clustering algorithm. This algorithm grouped the WSN to suitable sectors. This leads to balanced energy dissipation. To this end, the position of BS is placed at the centre of WSN field. Based on its central angle, the sequence number of its sector is assigned to each sensor node within circular field. The partition circle is rotated by BS for the next round, when data messages in a sector exceed other sectors. In this respect, the even energy consumption among the sensor nodes will be achieved.

5.2 Fault Management Based Approaches

The fault management system is implemented in WSNs to defeat any accidental fault may happen within entire network. Investigation in (Haoet al., 2010) introduces redundant CH nodes as a solution for the CHs failure. The redundant CHs algorithm is suitable for failure recovery, but no longer affects on monitoring and fault detection stages. In this context, the fault management system must be considered as complete platform.

Reference (Khan *et al.*, 2010) enhances the efficiency and reliability of WSN through the addition of functionalities like failure detection and failure recovery to behave as an integrated function of the sensor network toward managing systems of the clustered based protocols. This algorithm implements failure management platform by four levels, such as rotation of self-arranged cluster head sensor nodes CHs, detection of failure sensor node, detection of failure cluster head nodes and recovery function of failure cluster head node CH. However, this algorithm handles sensor network failure for all levels of clusters levels except Zone Manager (ZM) sensor node. To the best of our knowledge, the shortcoming of this algorithm is transmitting more messages regarding cluster members lists zone manager sensor devices. To this end, additional overheads are added like more memory usage and energy overheads. Furthermore, this algorithm no longer conducts any failure recovery scenario for faulty ZM that may affect the performance of the system.

Reference (Min *et al.*, 2010) implements a smart check pointing algorithm for handling detected failure and faulty CH nodes recovery. The contribution of this approach is to handle the transient faulty CHs nodes issues. The functionality of any CH node is to broadcast its routing and merged info to the existing backup sensor devices. This is to minimize the costs of recovery processes within entire WSN. Factors like number of backup sensor devices and check pointing interval are determined the level of algorithm reliability per cluster.

6. AGGREGATION STAGE OF SENSED DATA BASED APPROACHES

The redundancy is common phenomena in WSNs, based on numerous sensor devices are redundantly distributed on large area for data collecting purposes. The redundancy phenomena obligate a level of redundancy to accomplish a required QoS. In this respect, this phenomena imposes additional energy overhead within entire WSNs. Based on this, to keep the network work, data aggregation approaches are proposed. There are two existing surveys on sensed data aggregation

stage (Jesus *et al.*, 2011; Thangaraj and Punitha 2011). These surveys introduce several sensed data aggregation stage based algorithms within clustering routing protocols.

Reference (Nie and Li 2011) points out the significance of clustering based mechanism for Structural Health Monitoring (SHM) application based on large amount of processed data. The algorithm aim is to provide intelligent monitoring mechanism through arranging the data aggregation mechanism into a three-tier hierarchy. We can conclude that the outcome of this paper through three points, such as data processed by sensor node are concise and meaningful, the security of aggregated messages is significant and energy efficient algorithm.

To mitigate redundant messages within entire network, the paper (Wei *et al.*, 2011) conducts an aggregation technique based on prediction. On this context, this work minimizes the overhead based on correlation of temporal data. To the best our knowledge, it is significant to think about barter among messages overheads and limiting simultaneous errors to accomplish rational level of accuracy.

7. COMMUNICATION STAGE OF SENSED DATA BASED APPROACHES

Data communication phase is achieved by sending packets from normal cluster members toward their head clusters CH node within area of their intra-clusters. Then the clusters heads CHs nodes send their packets to the sink node which is assumed as BS within area of inter-cluster based on underlying application. In this section, we classify the existing algorithms in literature into two communication data; intra-cluster and inter-cluster based on hops number and sensor device type assigned to pass packets:

7.1 Intra –Cluster Communication Based Approaches

Reference (Nazir and Hasbullah 2010) handles the shortcoming of assigning 32 fixed Time Division Multiple Access (TDMA) slots to the normal nodes (NNs) within clusters by elected

CH nodes, conducted by the LEACH algorithm (Heinzelman *et al.*, 2000). In this respect, a dynamic TDMA schedule is presented, which based on traffic load changes. In this context, all nodes have variable slots time based on amount of data traffic in prior cycle. To the best of our knowledge, the contribution of this algorithm is based on assuming the load of current cycle to be used for the next cycles; this is no longer valid for various scenarios.

7.2 Inter -Cluster Communication Based Approaches

Multi Hop by Super Cluster Heads Based Approach: A novel election stage of SCH nodes is conducted by (Gawdan 2014). Each elected CH node sends its Residual Energy (RE), CH location and ID to the sink node which is BS. The BS computed E_A and D_{CH-BS} for clusters heads CHs devices. Consequently, the CH nodes are elected to become SCH nodes as long as conditions given in (1) and (2) are valid as mentioned in Section 3. This novel algorithm helps to achieve balanced energy of the load distribution in the entire network. After election stage of SCH nodes and assigned related clusters heads CHs nodes, sink node which is BS transmits "SCHs" packet within entire WSN which contains the ID_{SCH} nodes. When the ID_{SCH} matches with node identifier ID then that sensor node is assumed SCH device otherwise, the sensor node is remaining as cluster member for the current round. Based on our survey, this novel algorithm provides the reliability, functionality and energy aware approach upon hierarchically cluster based WSNs.

Multi Hop by Cluster Heads Based Approach: Reference (Farooq *et al.*, 2010) conducts an algorithm based on multi-layer clustering. In this context, cluster head distance to the upper level CH node is minimized. This algorithm proposes two types of CHs election techniques: distributed technique for low level cluster and centralized technique for high level cluster. To this respect, the sensor nodes within entire WSN connect to the BS which is sink node using their assigned low level of power. Concurrently, replied packets of CHs nodes are received only by sink node. To this end, the BS

node announces CHs-packet, includes list of IDs of the CHs nodes within first tier clusters, and normal sensor responds to CHs packet.

Consequently, this approach builds hierarchically cluster-based algorithm where any sensor node within given tier can access the sink node which is BS within same hops number, thus, saving amount of power in entire network to some extent.

Multi Hop by Relay Nodes (RNs) Based Approach: The algorithm presents in (Shenet *et al.*, 2010) assigns the task of path finding to CHs nodes, while data transmission functionality is given to gateway nodes. Within set up phase, the network is arranged into contagious overlapped clusters, and then the gateway sensor nodes are elected among all sensor devices within overlapped regions. Consequently, when any sensor device needs to send its data to sink device, the sensor device sends inform-message to its cluster head device which provides the requested info about optimum gateway sensor device. To the best of our knowledge, this algorithm is no longer mentioned any work methodology for factors and conditions used in gateways election phase and for path finding process as well.

8. PROPOSED TAXONOMY FOR POPULAR CLUSTER BASED ALGORITHMS IN WSNs

The current Survey classifies the popular existing cluster based algorithms in WSNs according to implemented methods into three main categories: chain, grid and block cluster based algorithms:

The popular chain cluster based algorithms are: (CCS: Jung *et al.*, 2007; PEGASIS: Lindsey *et al.*, 2002; TSC: Gautamet *et al.*, 2009).

The popular grid cluster based algorithms are: (TTDD: Luo *et al.*, 2005; GAF: Y. Xu *et al.*, 2001; PANEL: Buttyan and Schaffer 2007; SLGC: Deavaret *et al.*, 2012).

The popular block cluster based algorithms are: (CCM: Tang *et al.*, 2010; TEEN: Manjeshwar and Agrawal 2001; LEACH: Heinzelman 2000; LEACH-VF: Yasseinet *et al.*, 2009; EECS:

Ye *et al.*, 2005; UCS: Soro and Heinzelman 2005; HEED: Younis and Fahmy 2004).

Also this survey summarizes a brief comparison between popular existing algorithms on the basis of well-known factors as shown in Table 3.

Table 3. Comparison between popular clusters based algorithms

Algorithm	Efficient Energy	Latency	Stability	Scalability	Energy Balancing	Complexity Degree
LEACH	V. Low	V. Small	Fair	V. Low	Fair	Low
TTDD	V. Low	V. Large	High	Low	Good	Low
GAF	Fair	V. Small	Fair	High	Fair	Fair
CCM	V. Low	Small	High	Low	Fair	Fair
TEEN	High	Small	High	Low	Good	High
SLGS	Fair	V. Small	Fair	Low	Fair	Fair
PEGASIS	Low	V. Large	Low	Low	Fair	High

9. CONCLUSIONS AND SOME FUTURE STUDY ISSUES

9.1 Conclusions

The main goal of all proposed studies in literature is to improve the performance of the underlying application and increase the lifetime of WSNs. The existing schemes concentrated on enhancing the cluster based characteristics in a particular protocol work phase. This study has introduced a comprehensive survey upon most existing approaches in the field of cluster based issues for WSNs. This survey summaries the existing schemes contribution to any phase of clustered based work. The survey also offers the classification of the existing protocols in specific phase based on their goal and scenarios. Besides, this survey provides

necessary suggestions for many issues in each stage of the protocol work. Finally, this survey summarizes a brief comparison between popular existing algorithms on the basis of well-known factors as shown in Table 3. Based on this survey, we provide the following contributions as essential steps for improving the basic issues related to cluster based approaches:

The term of formation stage of clusters can implement resource aware protocols to process the cluster sizes for conducting balanced energy distribution within WSNs.

The powerful nodes which possess the highest energy level are more favorable and witnessed feature among other attributes for the election purpose.

For multi-hop scenarios, it is required to implement a temporary routing protocol in initial stages for large size WSNs. Generally, this adds additional overhead, affects on scalability and makes the algorithm more complex.

To enhance the time of transition in fault recovery cases, it is significant to put in mind fault management systems.

9.2 Future Study Issues

Based on the current survey results, there are several future open issues that need more investigations:

It is essential to remember that the recent optimum percentage number of CHs was 7% of the total alive nodes within entire WSNs. This value is determined in (Gawdan 2014) by calculation, energy consumption model and distances factors.

The term of scalability in the clustered based algorithms towards the additional devices must be carefully studied.

Management issues of transient fault as a result of the temporal link fault for clustering approaches need insight and more attention.

The demise of the sensor devices changes the network topology. So there is a strong need to put into consideration the robustness of clustering approaches towards the topology changes.

Proposed a taxonomy for a popular cluster based algorithms on the basis of implemented methods.

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